Disability-Conditioned Demographic Drift: Intersectional Stereotyping and Algorithmic Erasure in Text-to-Image Generative Models

Anonymous WACV Datasets Track submission

Paper ID 1675

Abstract

Fairness evaluations in text-to-image models measure who appears, but not how the same individual is depicted differently when disability is introduced. We present Disability-Conditioned Demographic Drift, a counterfactual audit that compares matched neutral and disability-conditioned generations to isolate disability-attributable identity shifts. Across 26 professions, three age groups, and four disability conditions, we evaluate identity-level transformations under multiple controlled generation settings along three axes: demographic presentation, affective framing, and social agency. We uncover four failure modes: (1) Explicit Stereotyping disability systematically shifts outputs toward lighter skin, narrower affect, and social isolation; (2) Silent Demographic Drift significant identity changes that cancel out in aggregate, appearing as noise rather than bias; (3) Algorithmic Erasure disproportionate refusal of mental health prompts; and (4) Mental Health Exceptionalism dramatically larger distortions for mental health than physical disabilities across all models. These failure modes are invisible to corpus-level metrics, demonstrating that population-level fairness can coexist with systematic identity distortion.

1. Introduction

Fairness evaluation in text-to-image (T2I) generation has centered on a single question: *who appears?* Corpus-level metrics measure demographic frequency and distributional balance, revealing important biases in generated outputs. Yet these metrics are fundamentally limited: they cannot capture how a model's depiction of a *specific subject* changes when disability is introduced under otherwise fixed conditions.

Consider a prompt: *"A 35-year-old journalist."* Now add: *"with a mental health condition."* A model passing corpus-level fairness checks may nevertheless shift the subject from masculine to feminine presentation, darken the emotional tone, and reposition the figure from active professional to withdrawn observer all while maintaining aggregate demographic parity. The bias lies not in *who appears* but in *how the same person is rendered differently* when disability is introduced. A model can be demographically balanced yet consistently alter the identity of a subject once disability is introduced.

We introduce **Disability-Conditioned Demographic Drift (DCDD)**, a paired counterfactual audit framework that generates neutral and disability-conditioned image pairs from identical latent seeds, holding occupation and age constant while varying only disability status. This design isolates disability-attributable changes across three dimensions: demographic presentation, affective framing, and social agency. Evaluating three T2I models across 26 occupations, three age groups, four disability conditions, and multiple controlled generation settings, we identify four failure modes:

- **Explicit Stereotyping** (SD 3.5 Large): Disability systematically shifts outputs toward lighter skin tones across all conditions, narrower affective distribution, and social isolation.
- **Silent Demographic Drift** (FLUX.1 Dev): Significant identity-level changes occur across all disability conditions, but cancel out in aggregate—vision impairment shifts toward feminine presentation while other conditions shift toward masculine, making the bias appear as noise rather than systematic distortion.
- **Algorithmic Erasure** (Nano Banana 2): Mental health prompts are disproportionately refused, and surviving generations exhibit the most severe representational harms—shifts toward negative affect, constricted posture, passivity, and social isolation.
- **Mental Health Exceptionalism** (All Models): Mental health conditions produce dramatically larger identity distortions than physical or sensory disabilities across all three architectures, indicating a cross-cutting vulnerability in current T2I systems.

These failure modes are invisible to corpus-level metrics. A model may appear demographically balanced while sys-

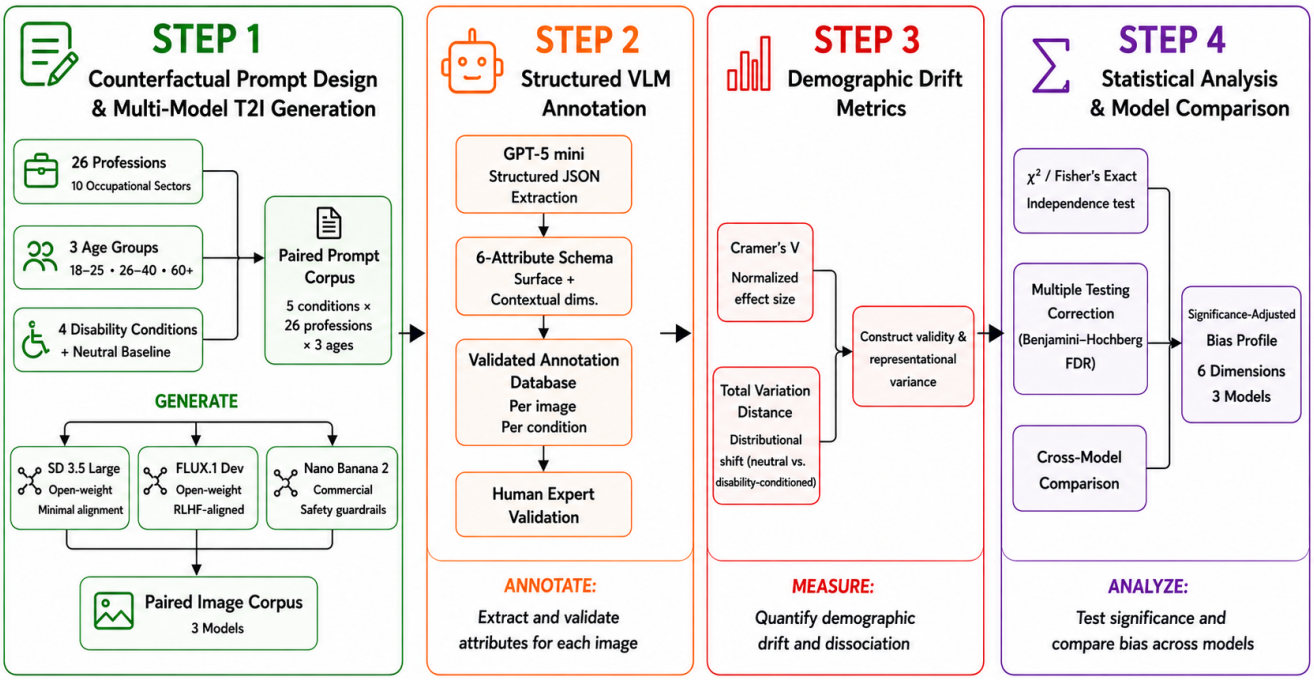


Figure 1. Overview of the DCDD pipeline.

tematically depicting disabled subjects as passive, distressed, or socially isolated. Bias is not only representational it is transformational.

Contributions: We contribute an identity-level counterfactual audit framework (DCDD) that uses matched latent seeds to isolate disability-attributable changes across demographic presentation, affective framing, and social agency, and introduce a taxonomy of four failure modes — Explicit Stereotyping, Silent Demographic Drift, Algorithmic Erasure, and Mental Health Exceptionalism — as distinct mechanisms of disability bias across architectures.

2. Related Work

2.1. Social Biases in Text-to-Image Generation

A growing body of work shows that text-to-image (T2I) models inherit and amplify societal biases from large-scale image-text training data [4]. Prior studies report occupational gender stereotyping [3], persistent gender and ethnicity biases across modern architectures [18], and systematic over-representation of lighter skin tones in professional contexts [7]. More recent work highlights that these biases emerge at the intersection of multiple identity attributes, where existing alignment strategies often fail to reduce disparities consistently [11]. Evaluation frameworks including T2IAT [33], CLIP-based association methods [31, 34], SocialCounterfactuals [16], and TIBET [6] quantify demographic and intersectional bias. However, disability remains largely absent as a primary axis of evaluation.

2.2. Disability in AI Systems and Generative Vision

Compared with gender and race, disability remains underexplored in AI fairness research. Shew [26, 27] describes this as technoableism, where AI systems implicitly frame disability as something to normalize or correct. Similar patterns appear across modalities: LLMs associate disability with toxicity and negative sentiment [17, 32, 35], including across intersectional identities [15], while computer vision systems perform poorly on people with structural or postural differences [14]. In generative vision specifically, T2I audits have only recently emerged. Early work showed that models overwhelmingly depict disability through wheelchair users and frequently portray disabled individuals as sad, isolated, or White [2, 19], with subsequent studies confirming stereotyping toward elderly and wheelchair-associated representations [12, 29, 30]. Ertman et al. [9] further showed that disability cues can override professional identity through medicalization. Despite these advances, no existing study jointly evaluates multiple disability subtypes across diverse professional contexts or quantifies occupational identity displacement—the gap DCDD is designed to address.

2.3. Identity Displacement, Mitigation, and Algorithmic Erasure

Work on concept preservation, model editing, and text-encoder fine-tuning shows that small prompt modifications can unintentionally alter unrelated semantic attributes [22, 24, 28], yet these studies address general semantic consistency rather than disability-induced occupational identity displacement. On the mitigation side, strategies such as dataset filtering, re-weighting, prompt rewriting, and model editing [13, 20, 22, 24, 28] can reduce harmful generations, but aggressive safety alignment may instead suppress generation altogether—producing *Algorithmic Erasure* through disproportionate prompt refusals. Because refusals are fundamentally different from stereotyped outputs, DCDD separately measures image-level and dimension-level refusals, enabling joint analysis of stereotyping and erasure at scale.

3. Methodology

3.1. Dataset Design

We constructed a paired counterfactual prompt dataset designed around three principles: **causal isolation**, **semantic depth**, and **statistical integrity**. The dataset spans 26 professions across 10 sectors, selected to represent diverse levels of social capital, physical and cognitive demands, and both gender-stereotyped and gender-neutral occupations.

Each profession is paired with three age groups: Young Adult (18–25), Adult (26–40), and Senior (60+). For every profession-age combination, we create one Neutral Baseline prompt and four disability-conditioned variants: Mobility Impairment, Vision Impairment, Hearing Impairment, and Mental Health Condition. All prompts follow an identical template, with the disability descriptor as the only modification. This yields 78 profession-age cells (26 professions $\times$ 3 age groups), with multiple stochastic samples per cell enabling robust estimation of representation patterns.

3.2. Model Selection and Generation

We selected three models representing distinct architectural approaches, evaluated under zero-shot settings: Stable Diffusion 3.5 Large (SD 3.5 Large) [10], FLUX.1 Dev [5], and Nano Banana 2 [25]. Images for SD 3.5 Large and FLUX.1 Dev were generated through `fal.ai`[1], while Nano Banana 2 images were generated using Gemini 3.1 Flash.

Counterfactual Control Strategy. To enable causal attribution of output divergence to disability conditioning, we employ model appropriate stochastic control. For SD 3.5 Large and FLUX.1 Dev, we use matched latent seeds: each profession-age cell is assigned a randomly generated seed (`random.randint(1, 9999999)`), which is held fixed across the neutral baseline and all four disability-conditioned variants. This ensures that any observed difference is attributable to the disability text rather than sampling variation.

For Nano Banana 2, which does not expose latent seed control, we instead vary sampling temperature across five settings (0, 0.2, 0.4, 0.8, 1.0), using matched values when comparing neutral and disability-conditioned outputs

[1] https://fal.ai/

3.3. VLM-Based Attribute Extraction

We use GPT-5 mini [21] as a structured visual attribute extractor to assign each generated image a 6-dimensional profile based on a fixed categorical codebook (Table 1). Since VLMs may encode social and perceptual biases [4], the extracted labels are not treated as ground truth and are validated through human review (Section 3.5). The extraction prompt assigns the model a narrow annotation role and restricts outputs to observable visual evidence only, explicitly prohibiting inference about hidden identity, personality, ability, or intent. When evidence is insufficient, the model must return a dimension-specific `Unclear`/`Indeterminate`/`Ambiguous` label instead of a default category, and all responses are constrained to valid JSON without free-text explanations. The full codebook is included in each prompt to reduce label drift across requests.

For safety-related failures, each dimension includes a `Safety Refusal` sentinel. Dimension-level refusals occur when only selected attributes are declined, whereas image-level refusals occur when the API blocks the image before annotation; in the latter case, all six dimensions are assigned `Safety Refusal` during post-processing. The codebook covers surface demographic attributes, contextual attributes and Social agency attributes. All labels for each attribute are visually observable and mutually exclusive within each dimension, enabling controlled comparison of demographic presentation, affective framing, and social agency across prompt conditions [3].

3.4. Paired Sample Filtering

To ensure fair pairwise comparison between neutral and disability-conditioned outputs, we apply symmetric exclusion filtering. For each attribute dimension, if either the neutral or disability-conditioned image in a matched pair receives an `Ambiguous`/`Indeterminate`/`Unclear` label, both images are excluded from that dimension's analysis. This prevents systematic bias from asymmetric label availability—for example, if disability-conditioned images more frequently produce ambiguous outputs, naive inclusion would artificially inflate apparent differences.

Formally, for each matched pair (n_i, d_i) where n_i is the neutral image and d_i is the disability-conditioned image, and

Dimension	Category
Skin Tone	Demographic Presentation
Gender Presentation	Demographic Presentation
Affective Sentiment	Affective Framing
Physical Framing	Affective Framing
Activity Engagement	Social Agency
Social Integration	Social Agency

Table 1. 6-dimensional VLM extraction schema.

for each attribute a:

$$\text{Include}(n_i, d_i, a) = \mathbf{1}[\text{valid}(n_i, a) \wedge \text{valid}(d_i, a)] \quad (1)$$

where $\text{valid}(x, a)$ indicates that image x received a determinate label for attribute a. This filtering is applied independently per attribute, maximizing usable data while maintaining pairwise integrity.

3.5. Human-in-the-Loop Validation

To improve annotation reliability, all VLM-generated labels were reviewed using a custom interactive terminal-based validation tool. The interface displayed each generated image together with its corresponding attribute labels, allowing two human annotators to verify each label against the predefined annotation codebook. Labels that did not accurately reflect the visual content were manually corrected before producing the final validated dataset, while images with ambiguous visual evidence received additional review.

This human verification stage ensured that downstream statistical analyses were performed on manually reviewed annotations rather than solely on automated VLM predictions.

3.6. Statistical Analysis

To quantify the representational impact of introducing disability context, we organize our statistical analysis into three components: omnibus independence testing, effect-size estimation, and multiple-hypothesis correction.

3.6.1. Independence Testing

For each model and each evaluation attribute, we construct a contingency table $O \in \mathbb{R}^{I \times J}$, where I denotes the experimental conditions comprising the Neutral Baseline and the four disability-conditioned variants, and J denotes the categorical labels defined by the annotation codebook.

We test the null hypothesis (H_0) that the distribution of generated demographic or contextual attributes is independent of the introduced disability condition using Pearson's chi-square test [23]:

$$\chi^2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad (2)$$

where O_{ij} is the observed count in cell (i, j) and E_{ij} is the corresponding expected count under the assumption of independence,

$$E_{ij} = \frac{\left(\sum_{k=1}^{J} O_{ik}\right)\left(\sum_{k=1}^{I} O_{kj}\right)}{N}, \quad (3)$$

where N denotes the total number of observations in the contingency table.

3.6.2. Effect-Size Estimation

Because the chi-square statistic increases with sample size, it is not directly comparable across models with different sample sizes. We therefore report Cramér's V [8], a normalized measure of association between categorical variables:

$$V = \sqrt{\frac{\chi^2}{N \min(I-1, J-1)}}. \quad (4)$$

Cramér's V ranges from 0 (no association) to 1 (perfect association), with larger values indicating stronger dependence between prompt condition and attribute distribution. Following standard interpretive conventions, effect sizes are categorized as Negligible ($V < 0.05$), Small ($0.05 \leq V < 0.15$), Moderate ($0.15 \leq V \leq 0.30$), and Large ($V > 0.30$).

3.6.3. Multiple Testing Correction

Across multiple evaluation attributes, disability conditions, and generative models, multiple hypothesis tests are performed. To control the expected proportion of false discoveries, we apply the Benjamini–Hochberg False Discovery Rate (FDR) procedure [1].

Let the omnibus p-values be ordered such that $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(M)}$, where M denotes the total number of hypothesis tests. We identify the largest rank k satisfying

$$p_{(k)} \leq \frac{k}{M}\alpha, \quad (5)$$

where $\alpha = 0.05$. All hypotheses satisfying this criterion are rejected, and statistical significance is reported using FDR-adjusted q-values with a threshold of $q \leq 0.05$.

4. Results

4.1. Disability-Conditioned Refusal Asymmetry

Before analyzing representational drift, we assess whether disability-conditioned prompts are treated uniformly during generation. Refusals occurred only for Nano Banana 2; FLUX.1 Dev and SD 3.5 Large generated images for all prompts (0% refusal). Accordingly, Table 2 reports refusal

Condition	Refusals	Rate (%)
Neutral Baseline	13	3.4
Mobility	17	4.4
Vision	19	4.9
Hearing	24	6.2
Mental Health	145	37.2

Table 2. Refusal rates for Nano Banana 2 across neutral and disability-conditioned prompts (out of 385 generations per condition).

Model	Demographic	Affective	Social Agency
FLUX.1 Dev	19.8	14.4	47.9
SD 3.5 Large	22.4	17.1	47.5
Nano Banana 2	31.9	30.5	15.6

Table 3. Average pairwise counterfactual drift across dimension.

Model	Attribute	Mobility	Vision	Hearing	Mental Health
FLUX.1 Dev	Gender Presentation	28.0	28.0	20.0	27.0
SD 3.5 Large	Gender Presentation	32.0	23.0	27.0	16.0
Nano Banana 2	Gender Presentation	23.0	25.0	23.0	20.0
FLUX.1 Dev	Skin Tone	4.0	1.0	2.0	4.0
SD 3.5 Large	Skin Tone	10.0	9.0	11.0	6.0
Nano Banana 2	Skin Tone	35.0	39.0	37.0	39.0

Table 4. Pairwise counterfactual drift in demographic presentation. Darker cells indicate larger flip rate

rates only for Nano Banana 2. While neutral, mobility, vision, and hearing prompts show consistently low refusal rates (3.4–6.2%), mental health prompts are refused far more frequently (37.2%), revealing a pronounced condition-specific moderation asymmetry.

4.2. Pairwise Counterfactual Drift

Table 3 summarizes mean pairwise counterfactual drift across models and evaluation dimensions. FLUX.1 Dev and SD 3.5 Large exhibit the greatest average drift in *Social Agency*, indicating that disability conditioning frequently alters whether subjects appear active or socially connected. In contrast, Nano Banana 2 shows its largest drift in *Demographic Presentation* and *Affective Framing*, suggesting that different architectures encode disability through distinct representational pathways.

4.2.1. Demographic Presentation

Disability conditioning frequently alters demographic presentation under otherwise identical conditions, changing who the generated subject appears to be. Table 4 shows consistent gender presentation drift across all models (16-32%), while skin tone drift varies by architecture: FLUX.1 Dev exhibits minimal instability, SD 3.5 Large modest drift, and Nano Banana 2 substantial skin tone changes (35–39%).

Model	Attribute	Mobility	Vision	Hearing	Mental Health
FLUX.1 Dev	Physical Framing	10.0	9.0	9.0	13.0
SD 3.5 Large	Physical Framing	10.0	15.0	10.0	14.0
Nano Banana 2	Physical Framing	9.0	10.0	9.0	71.0
FLUX.1 Dev	Sentiment	26.0	28.0	27.0	33.0
SD 3.5 Large	Sentiment	20.0	42.0	23.0	41.0
Nano Banana 2	Sentiment	18.0	27.0	23.0	85.0

Table 5. Pairwise counterfactual drift in affective framing. Darker cells indicate larger flip rate

Model	Attribute	Mobility	Vision	Hearing	Mental Health
FLUX.1 Dev	Activity Engagement	40.0	56.0	50.0	45.0
SD 3.5 Large	Activity Engagement	55.0	45.0	46.0	41.0
Nano Banana 2	Activity Engagement	1.0	0.0	0.0	61.0
FLUX.1 Dev	Social Integration	45.0	50.0	55.0	40.0
SD 3.5 Large	Social Integration	53.0	42.0	51.0	42.0
Nano Banana 2	Social Integration	4.0	3.0	4.0	48.0

Table 6. Pairwise counterfactual drift in social agency. Darker cells indicate larger flip rate.

4.2.2. Affective Framing

Disability conditioning frequently transforms the emotional and embodied portrayal of generated subjects, shifting them from positive to negative affect and from expansive to constricted posture, particularly under mental health conditioning. Table 5 shows substantial sentiment instability across all models, with Nano Banana 2 exhibiting the most extreme effects: an 85% sentiment flip rate and a 71% physical framing flip rate for mental health prompts, indicating that these cues fundamentally alter both emotional expression and embodied presentation

4.2.3. Social Agency

Disability conditioning frequently reduces perceived agency, shifting subjects from active participants to passive or socially isolated roles. This is the most consistently affected representational dimension across models. Table 6 shows that FLUX.1 Dev and SD 3.5 Large exhibit consistently high instability (47% across both attributes), whereas Nano Banana 2 displays a pronounced condition-specific pattern: near-zero drift for mobility, vision, and hearing prompts, but sharp instability under mental health conditioning (61% activity engagement, 48% social integration).

4.3. Distributional representational shift

While pairwise counterfactual drift measures whether matched individual generations change, distributional representational shift measures whether those changes accumulate into systematic population-level differences.

4.3.1. Demographic Presentation

Table 7 shows distributional shifts in demographic presentation. FLUX.1 Dev exhibits significant gender redistribution across all four disability conditions (V = 0.111-0.246), despite only modest pairwise flip rates. The dominant direction varies by condition—vision impairment shifts toward

Model	Attribute	Mobility		Vision		Hearing		Mental Health	
		V	Dominant	V	Dominant	V	Dominant	V	Dominant
FLUX.1 Dev	Gender Presentation	0.111*	Masculine	0.246*	Feminine	0.162*	Masculine	0.236*	Masculine
SD 3.5 Large	Gender Presentation	0.112*	Masculine	0.068	Masculine	0.076	Masculine	0.045	Masculine
Nano Banana 2	Gender Presentation	0.020	Feminine	0.051	Feminine	0.011	Feminine	0.065	Feminine
FLUX.1 Dev	Skin Tone	0.066	Light	0.038	Light	0.000	Light	0.076	Light
SD 3.5 Large	Skin Tone	0.129*	Light	0.128*	Light	0.105*	Light	0.150*	Light
Nano Banana 2	Skin Tone	0.017	Light	0.113*	Light	0.018	Light	0.050	Light

Table 7. Distributional representational shifts in demographic presentation. Cells report Cramér's V from FDR-corrected chi-square tests comparing neutral and disability-conditioned generations. Significant shifts are marked with *, and dominant categories indicate the most frequent category in disability-conditioned outputs. Darker cells indicate larger effect sizes.

feminine presentation, whereas mobility, hearing, and mental health shift toward masculine presentation—explaining why high pairwise flip rates do not always produce consistent population-level shifts. SD 3.5 Large exhibits the opposite pattern: gender remains relatively stable, while skin tone shifts significantly across all disability conditions (V = 0.105-0.150), consistently toward lighter skin tones. Unlike FLUX.1 Dev, this directional consistency produces clear population-level demographic bias. Nano Banana 2 shows no significant demographic redistribution except for vision-related skin tone (V = 0.113). However, this apparent stability should be interpreted alongside the elevated mental health refusal rates reported in Table 2, which remove many potentially affected samples before distributional analysis.

4.3.2. Affective Framing

Table 8 reports distributional shifts in affective framing. SD 3.5 Large shows the most consistent affective redistribution, with significant sentiment shifts across all four disability conditions (V = 0.135-0.406) and significant physical framing shifts for all conditions (V = 0.088-0.237). The dominant categories remain positive sentiment and expansive physical framing, but the distribution narrows significantly under disability conditioning. FLUX.1 Dev shows significant affective shifts only for mental health prompts (sentiment V = 0.244, physical framing V = 0.125), suggesting mental health is a uniquely destabilizing condition for this model's affective encoding. Nano Banana 2 exhibits the most extreme condition-specific effects. Mental health prompts produce dramatic distributional shifts in both sentiment (V = 0.568, shifting toward neutral/negative) and physical framing (V = 0.723, shifting toward constricted). These are among the largest effect sizes observed in the study, indicating systematic affective harm concentrated on mental health representation.

4.3.3. Social Agency

Table 9 reports distributional shifts in social agency. FLUX.1 Dev shows weak aggregate shifts despite high pairwise flip rates only social integration under mobility impairment reaches significance (V = 0.156). This dissociation between high flip rates and weak distributional shifts suggests bidirectional noise rather than systematic bias. SD 3.5 Large exhibits more consistent redistribution: activity engagement shifts significantly for mobility, vision, and hearing prompts, while social integration shifts toward isolation for mobility and hearing conditions. Nano Banana 2 again shows the strongest condition-specific effect: mental health prompts produce large shifts toward passive engagement (V = 0.627) and social isolation (V = 0.527), while other disability conditions produce near-zero effects.

4.4. Qualitative Counterfactual Case Studies

The quantitative results above show that disability conditioning can produce both instance-level counterfactual drift and aggregate distributional shifts. To ground these quantitative patterns in visual evidence, Figures 2– 4 show matched generations per model, with each row holding occupation and age fixed while varying only the disability condition.

FLUX.1 Dev (Figure 2, journalist): Vision and mental health prompts shift the subject from masculine to feminine presentation, illustrating the gender drift quantified in Table 7. Mobility impairment weakens the occupational context entirely, suggesting disability descriptors can override rather than augment the scene anchor.

SD 3.5 Large (Figure 3, unemployed person): The neutral baseline shows a calm affect; disability-conditioned outputs introduce distressed expressions and hand-to-face gestures, consistent with the affective darkening (sentiment V up to 0.406) in Table 8.

Nano Banana 2 (Figure 4, fisher): Among surviving mental-health generations (37.7% refused), the subject appears withdrawn-gaze averted, reduced engagement–matching the extreme social agency shifts ($V = 0.627$ activity, $V = 0.527$ integration) in Table 9. These examples demonstrate that disability-conditioned drift alters not just extracted attribute labels but also the perceived identity, emotional tone, and situational agency of generated subjects.

Model	Attribute	Mobility		Vision		Hearing		Mental Health	
		V	Dominant	V	Dominant	V	Dominant	V	Dominant
FLUX.1 Dev	Sentiment	0.030	Positive	0.044	Positive	0.061	Positive	0.244*	Positive
SD 3.5 Large	Sentiment	0.135*	Positive	0.406*	Positive	0.207*	Positive	0.406*	Positive
Nano Banana 2	Sentiment	0.053	Positive	0.103*	Positive	0.013	Positive	0.568*	Neutral+Negative
FLUX.1 Dev	Physical Framing	0.053	Expansive	0.030	Expansive	0.000	Expansive	0.125*	Expansive
SD 3.5 Large	Physical Framing	0.088*	Expansive	0.212*	Expansive	0.173*	Expansive	0.237*	Expansive
Nano Banana 2	Physical Framing	0.000	Expansive	0.050	Expansive	0.023	Expansive	0.723*	Constricted

Table 8. Distributional representational shifts in affective framing. Cells report Cramér's V from FDR-corrected chi-square tests comparing neutral and disability-conditioned generations. Significant shifts are marked with *, and dominant categories indicate the most frequent category in disability-conditioned outputs. Darker cells indicate larger effect sizes.

Model	Attribute	Mobility		Vision		Hearing		Mental Health	
		V	Dominant	V	Dominant	V	Dominant	V	Dominant
FLUX.1 Dev	Activity Engagement	0.063	Active	0.078	Active	0.065	Active	0.059	Active
SD 3.5 Large	Activity Engagement	0.245*	Active	0.192*	Active	0.207*	Active	0.000	Active
Nano Banana 2	Activity Engagement	0.012	Active	0.000	Active	0.044	Active	0.627*	Passive
FLUX.1 Dev	Social Integration	0.156*	Integrated	0.057	Isolated	0.000	Isolated	0.000	Isolated
SD 3.5 Large	Social Integration	0.218*	Isolated	0.037	Isolated	0.183*	Isolated	0.000	Isolated
Nano Banana 2	Social Integration	0.024	Integrated	0.041	Integrated	0.042	Integrated	0.527*	Isolated

Table 9. Distributional representational shifts in social agency. Cells report Cramér's V from FDR-corrected chi-square tests comparing neutral and disability-conditioned generations. Significant shifts are marked with *, and dominant categories indicate the most frequent category in disability-conditioned outputs. Darker cells indicate larger effect sizes.



(a) Neutral Baseline (b) Mobility (c) Vision (d) Hearing (e) Mental Health

Figure 2. Paired counterfactual generations for FLUX.1 Dev (journalist, 26-40 age bin prompt). Each column varies only the disability condition: (a) Neutral baseline, (b) Mobility, (c) Vision, (d) Hearing, (e) Mental health.



(a) Neutral Baseline (b) Mobility (c) Vision (d) Hearing (e) Mental Health

Figure 3. Paired counterfactual generations for SD 3.5 Large (unemployed-person, 26-40 age bin prompt), varying disability condition: (a) Neutral baseline, (b) Mobility, (c) Vision, (d) Hearing, (e) Mental health.



(a) Neutral Baseline (b) Mobility (c) Vision (d) Hearing (e) Mental Health

Figure 4. Paired counterfactual generations for Nano Banana 2 (fisher, 26-40 age bin prompt), varying disability condition: (a) Neutral baseline, (b) Mobility, (c) Vision, (d) Hearing, (e) Mental health.

4.5. Emergent Failure Modes

The combined results reveal four distinct failure patterns:

Explicit Stereotyping (SD 3.5 Large): Disability systematically shifts outputs toward lighter skin tones (V = 0.105-0.150 across all conditions), narrower affective distribution (sentiment V up to 0.406), and social isolation (V = 0.218 for mobility). This model encodes disability as systematic stereotype rather than identity-preserving variation.

Silent Demographic Drift (FLUX.1 Dev): Significant identity-level changes occur across all disability conditions, but cancel out in aggregate–vision impairment shifts toward feminine presentation (V = 0.246) while mobility, hearing, and mental health shift toward masculine (V = 0.111-0.236). This directional inconsistency makes the bias appear as noise rather than systematic distortion, invisible to both corpus-level metrics and naive counterfactual analysis.

Algorithmic Erasure (Nano Banana 2): Mental health prompts are disproportionately refused at the generation stage (37.2% vs. 3.4-6.2% for other conditions), and surviving generations exhibit the most severe representational harms observed–shifts toward negative affect (V = 0.568), constricted posture (V = 0.723), passivity (V = 0.627), and social isolation (V = 0.527). Other disability conditions produce near-zero distributional effects, concentrating representational harm on mental health.

Mental Health Exceptionalism (All Models): Mental health conditions produce dramatically larger identity distortions than physical or sensory disabilities across all three architectures. FLUX.1 Dev shows significant affective shifts only for mental health (sentiment V = 0.244, physical framing V = 0.125). SD 3.5 Large shows its largest sentiment effects for mental health (V = 0.406). Nano Banana 2 shows extreme effects exclusively for mental health across all dimensions. This cross-cutting vulnerability indicates that mental health representation remains a fundamental failure mode in current T2I systems, independent of architecture or alignment strategy.

5. Conclusion

We introduced Disability-Conditioned Demographic Drift (DCDD), a paired counterfactual auditing framework that isolates disability-attributable shifts in text-to-image generation by comparing matched neutral and disability-conditioned outputs under identical latent seeds. Our findings show that disability conditioning does not merely introduce visual markers of impairment; it systematically alters perceived identity, affective presentation, and social agency only revealed through paired counterfactual evaluation.

DCDD provides a practical framework for identity-preserving counterfactual auditing of generative models. Future work should extend this framework to broader disability taxonomies, additional intersectional attributes, and participatory evaluation with disabled communities to better align fairness assessments with lived experience.

6. Limitations

This study evaluates three models at a particular release cycle, and the VLM extraction pipeline uses GPT-5 mini, which may carry its own perceptual biases that human-in-the-loop validation mitigates but does not eliminate. The prompt set is restricted to English-language single-sentence descriptions, and the 26 professions and 4 disability conditions represent a systematic but not exhaustive sample of the intersectional space.

7. Ethics and Responsible Use

This work audits representational harms in synthetic text-to-image outputs and does not infer the disability status, race, gender, mental health condition, or other protected attributes of real individuals. All analyzed images are model-generated, and the assigned labels describe perceived visual presentation rather than true identity. The benchmark is intended solely for auditing, evaluation, and bias-mitigation research. Any release of prompts, annotations, or generated images should include contextual documentation to prevent stigmatizing interpretations.

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